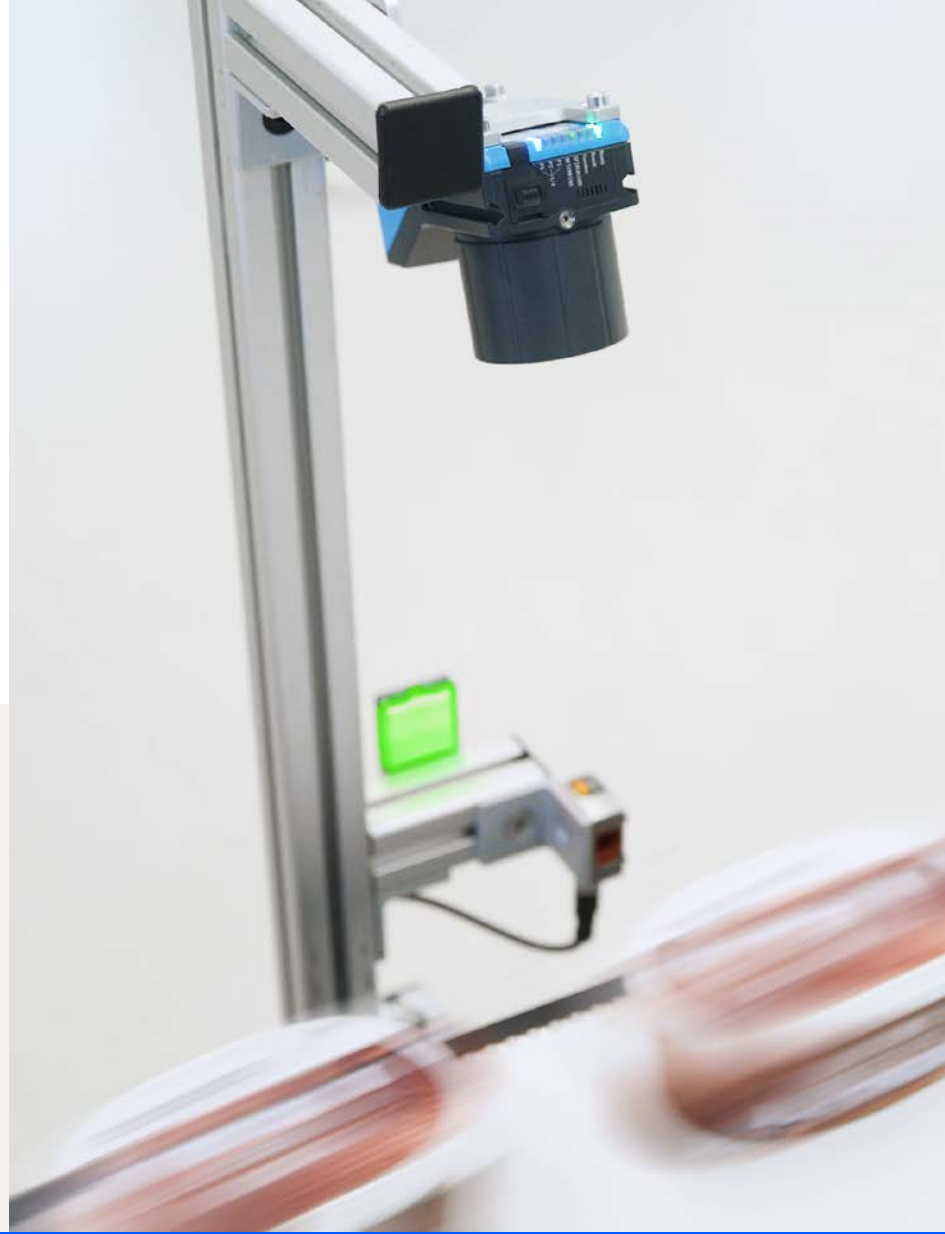


SICK.COM



OVERVIEW OF THE PRODUCTS

MACHINE VISION

2D machine vision, 3D machine vision, Sensor Integration Machine

SICK

Sensor Intelligence



WE GIVE YOU THE VISION TO PERFORM

Production and logistics teams need high-performing and easy-to-use machine vision solutions to drive ever-more dynamic automated operations. Being able to achieve the right vision configuration and progress rapidly to completely reliable operation is vital. Grounded in a history of pioneering technology, the wide-ranging 2D and 3D portfolio from SICK remains at the cutting edge of machine vision development. Supported by our global experts and benefiting from extensive customer application experience, SICK gives you the vision to perform.

Alongside world-class streaming cameras and easy-to-use vision sensors, the unique machine vision foundation software and digital services from SICK are focused on simplicity for our customers. Positioning, inspection, measuring and code-reading, whatever the combination, you are empowered to adapt and fast-track your application to optimum operating performance.

Our commitment to a common user experience gives you the confidence to grow, and the agility to adapt quickly to your product and process changes.

From tiny, intelligent sensors to high-performance 3D streaming cameras and Sensor Integration Machines, the broad portfolio of state-of-the-art hardware from SICK is supported by two unique software offers. Thanks to the SICK Nova foundation software, you get solution-oriented toolsets, ready to run directly onboard our vision sensors - solving your application in just minutes. You don't even need programming skills. Just access the toolset you need, including options for AI Deep Learning or configure your own application-specific solutions. When you need to stream high quality data, the intuitive but yet powerful Stream configuration software offers rapid integration into your own software ecosystem.

Learn more

sick.com/Machine_Vision



YOUR ADVANTAGES WITH MACHINE VISION FROM SICK



TECHNOLOGY - OPEN AND SCALABLE

From individual sensors to complete automation solutions, SICK adapts each project to your specific requirements.



INCREASE EFFICIENCY AND THROUGHPUT AND BOOST YOUR YIELD

Increased production uptime is achieved by detecting and correcting errors in real time, even in high-speed processes - for low failure rates, high throughput and on-time delivery.



GREATER FLEXIBILITY

State-of-the-art vision solutions can be used in a wide range of production processes and can be easily adapted to changes in series production.



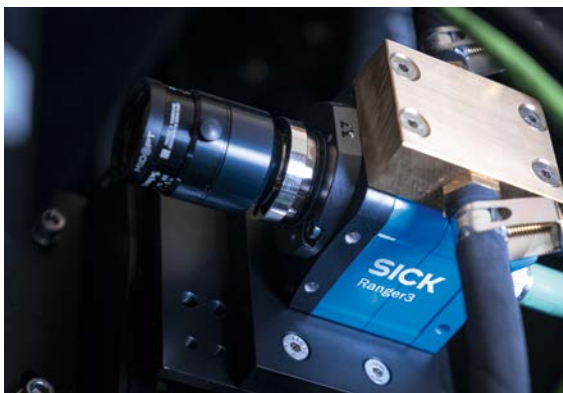
EASY TO OPERATE

Thanks to AI technologies, even demanding machine vision applications can be implemented effortlessly. The user-friendly configuration and operation of SICK Nova requires no software installation and enables simple integration.



2D MACHINE VISION

- SICK offers a powerful vision sensor portfolio designed to manage challenges in all industries where a standard sensor would not work. These vision sensors provide a full toolset for positioning, inspection, measurement and reading, depending on the variant. A flexible optical design fulfills the needs of almost all applications. Simplicity is ensured by automatic setup, intelligent algorithms and a intuitive web based user interface.



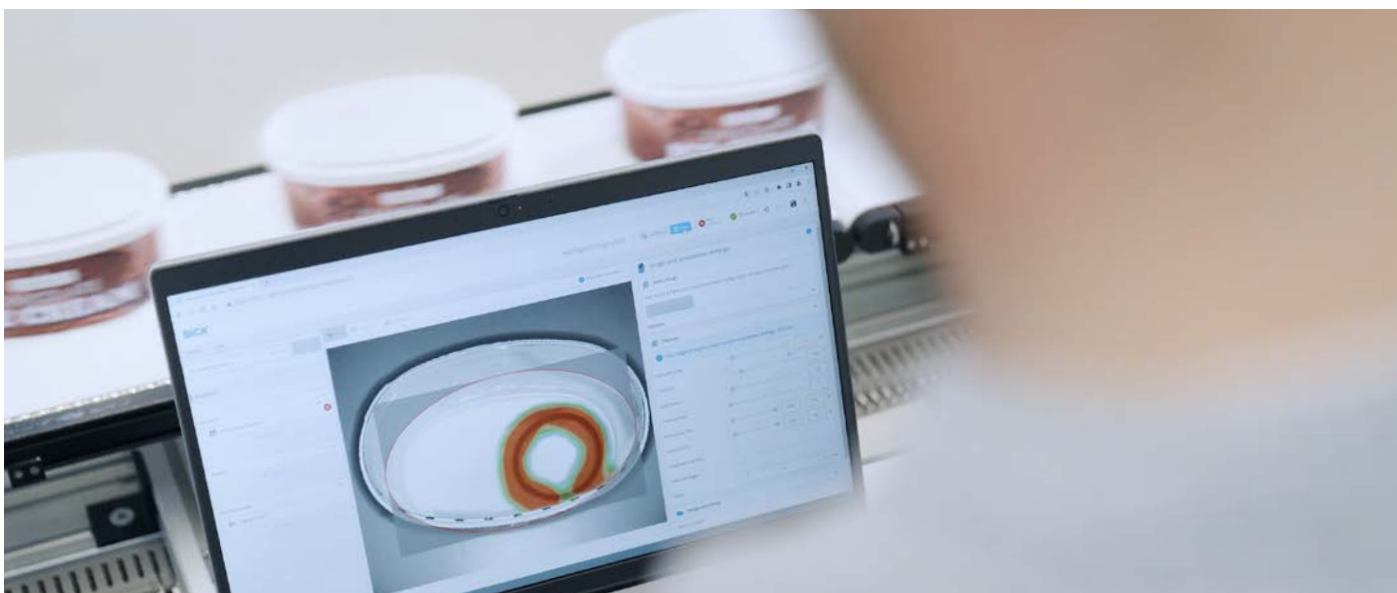
3D MACHINE VISION

- SICK's 3D vision series offers a wide range of powerful and flexible products designed for reliable operation in harsh industrial environments. They range from versatile high-speed cameras that deliver high quality 3D and contrast images to smart and configurable stand-alone sensors that facilitate rapid development and easy integration. Their scalability ensures a perfect fit with your 3D vision application.



MACHINE VISION SOFTWARE

- With simplicity for our customers in mind, SICK offers machine vision software solutions that seamlessly integrates with all vision sensors in our portfolio. Thanks to scalability in terms of functionality and integration into ecosystems, there is a solution for every requirement. When combined with the power of AI Deep Learning, allowing teaching by examples instead of setting up rules manually, machine vision truly becomes easy to use.





SICK NOVA

MACHINE VISION MADE EASY

Learn more
sick.com/SICK_Nova
 →



SICK Nova is the foundation software for configurable machine vision solutions in a broad array of industries and quality control applications. Featuring a user-friendly web interface and AI capabilities, SICK Nova supports experts and non-experts alike so they can quickly and easily handle even the most challenging applications.

SICK Nova runs on a multitude of 2D and 3D vision sensors as well as LiDAR sensors, leveraging different imaging technologies. Moreover, users can conveniently extend the software's functionality with both user-developed and ready-made SICK Nova plug-ins. With this kind of scalability, SICK Nova is a safe choice that ensures cost-effective ownership.

Scalable,
 on-device and
 easy-to-use





EASE OF USE

SICK Nova offers a user-friendly web interface with point-and-click configuration and embedded assistance, making the software quick and convenient to use.



SCALABILITY

SICK Nova runs seamlessly on SICK 2D and 3D vision sensors and has the same familiar user experience. With three toolset levels for users to choose from, including tools powered by AI deep learning, SICK Nova delivers a suitable solution for every need.



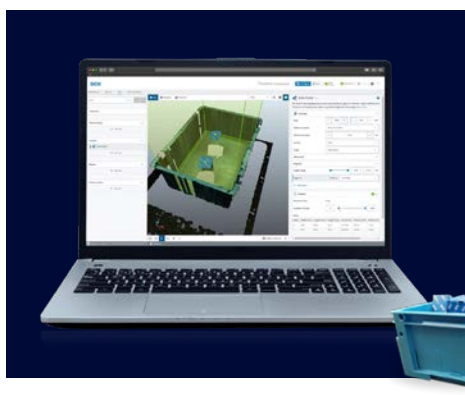
PLUG-INS

Users can expand the software's functionality using standard or custom SICK Nova plug-in tools available in the SICK AppPool. Users with a SICK AppSpace license can also develop customized plug-ins if needed.



AI CAPABILITIES

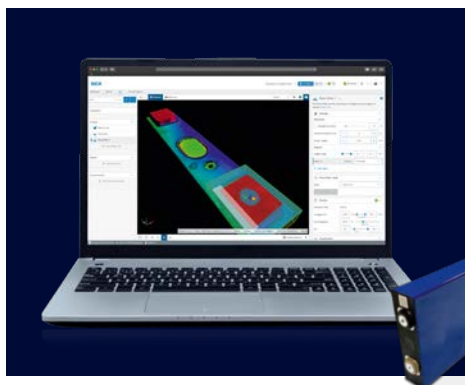
Powered by AI deep-learning technology, SICK Nova offers both on-device and cloud-based tools for complex object and defect detection, positioning, sorting and counting



PRESENCE INSPECTION



→ The Presence Inspection toolset provides solutions to a wide range of applications such as presence and absence detection, completeness checks, counting, sorting and fill level monitoring tasks. It comes with a fundamental, yet comprehensive, set of image analysis and communication tools that are easily configured using an intuitive graphical user interface in a web browser.



QUALITY INSPECTION



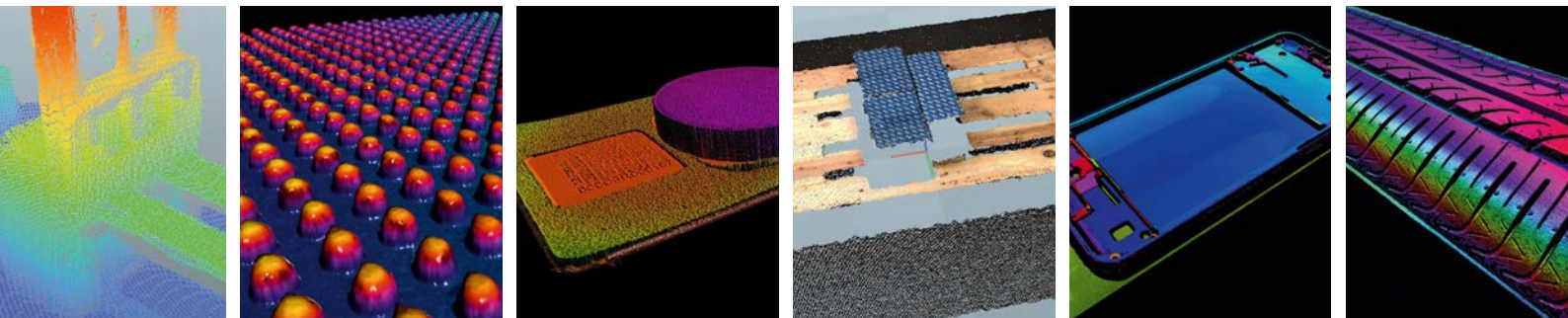
→ The Quality Inspection toolset enables solutions for detailed quality control, object localization and measuring in manufacturing and assembly verification applications. It can be used to ensure that produced items have the exact qualities required regarding presence, location and dimensions. In addition to all tools available in Presence Inspection toolset, the Quality Inspection toolset also includes a vast range of tools for quality assurance, positioning, dimensioning, calibration and identification.



INTELLIGENT INSPECTION



→ With the AI-powered Intelligent Inspection toolset, machine vision truly becomes easy to use! By teaching the vision sensor using examples instead of manually setting up rules, users can solve applications quickly while gaining new levels of inspection capabilities. Combining the benefits of AI tools and rule-based tools opens up extensive possibilities for solutions, regardless of how experienced the user is with machine vision. All traditional rule-based machine vision software tools from Quality Inspection toolset are included.



TAKE CONTROL OF YOUR 3D STREAMING DATA

SICK STREAM SOFTWARE SOLUTIONS

To easily handle streaming high-quality data, SICK offer configuration softwares that enables easy parameterization and rapid integration into your own software ecosystem.



STREAM SETUP

→ The [Stream Setup configuration software](#) comes included with the Ranger3 and Ruler3000 3D cameras and provides quick and easy configuration, collection, optimization and evaluation with the user-friendly processing tools for e.g., noise reduction and false data removal. With Stream SDK (software Development Kit), rapid camera integration is enabled by GigE vision and supportive C#/C++ APIs into your PC application.



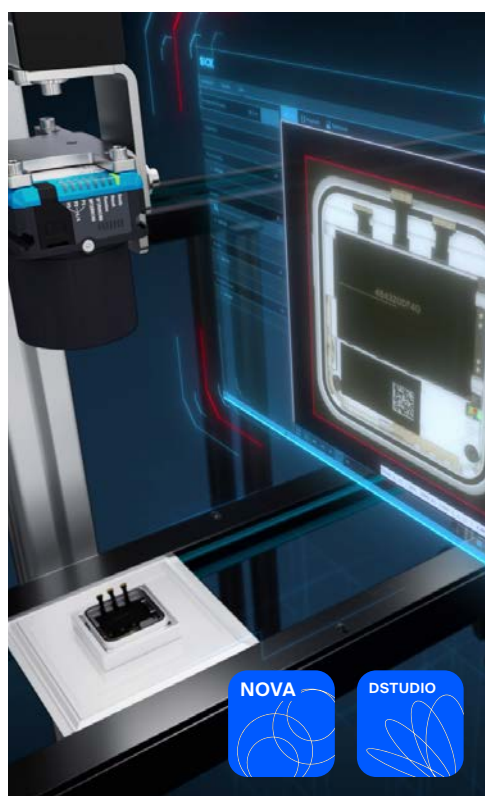
SOPAS ENGINEERING TOOL

→ The [SOPAS Engineering Tool](#) allows for easy configuration of the streaming products Visionary-T Mini CX and Visionary-S CX. It provides a quick and easy way to visualize and record data. Along with the SOPAS Engineering Tool, a comprehensive SDK enables a fast camera integration into your application by providing e.g. C++ and Python samples as well as ROS driver.





AI functionality makes machine vision tasks easier than ever. By teaching the vision sensor using examples instead of manually setting up rules, users can solve applications quickly while gaining new levels of inspection capabilities. Combining the benefits of AI tools and rule-based tools opens up extensive possibilities for solutions, regardless of how experienced the user is with machine vision.



WHY CHOOSE AI FROM SICK?



DEEP LEARNING MADE ACCESSIBLE

Configure your Deep Learning application with easy-to-use software tools. By teaching with examples instead of manually setting up rules, applications can be solved rapidly also by non-experts. In addition, the deep-learning-powered Intelligent Inspection toolset provides solutions to more complex defect detection and object classification applications where traditional rule-based tools may be insufficient on their own.



ON-DEVICE OR SICK dSTUDIO BASED TRAINING

Fast and easy on-device application buildup with the Intelligent Inspection toolset as well as optimized accuracy and execution speed with training in SICK dStudio.



EASY TO EXPAND AND CUSTOMIZE

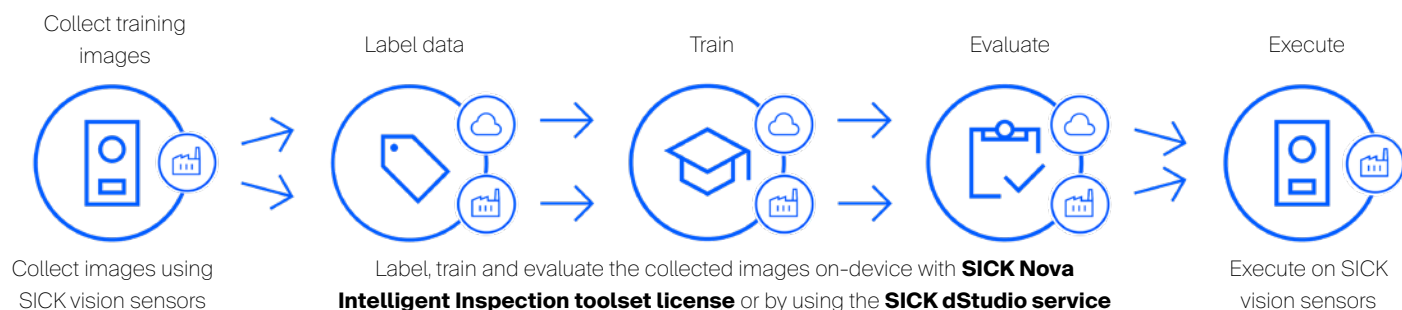
Our flexible product portfolio can easily be tailored to specific requirements, by yourself or with our experienced support teams. With SICK Nova Plug-in support it's easy to add-on functionality.



DEEP LEARNING DEDICATED SUPPORT AND SERVICE

Deep Learning experts specialized in your industry. Located where you are.

FAST AND EASY ON-DEVICE APPLICATION BUILDUP.
FOR OPTIMIZED ACCURACY AND EXECUTION SPEED, LABELLING, TRAINING AND EVALUATION CAN BE DONE WITH SICK DSTUDIO.



MACHINE VISION

PRODUCT		METHOD OF DATA PROVIDING				MAIN TASKS							
		Embedded image processing	Streaming	SICK AppSpace	SICK Nova	Classification	Identification	Collision warning / Navigation	Positioning	Presence detection	Quality Control	Measuring	Color inspection

2D MACHINE VISION

	InspectorP61x	●		●	●	●	●		●	●	●	●	●
	InspectorP62x	●		●	●	●	●		●	●	●	●	
	InspectorP63x	●		●	●	●	●		●	●	●	●	
	Inspector83x	●		●	●	●	●		●	●	●	●	●
	Inspector85x	●		●	●	●	●		●	●	●	●	
	picoCam2		●	1)	1)	1)	1)			1)	1)	1)	●
	midiCam2		●	1)	1)	1)	1)			1)	1)	1)	●

3D MACHINE VISION

	TriSpector1000	●		●					●	●	●	●	
	Ranger3		●						2)	2)	2)	2)	●
	Ruler3000		●		1)		1)		1)	1)	1)	1)	
	Visionary-S	●	3)	4)	4)			4)	4)	4)	4)	4)	●
	Visionary-T Mini	●	3)	4)	4)			4)	4)	4)	4)	4)	
	Visionary-B Two	●	●	●		●		●	●	●		●	●

SENSOR INTEGRATION MACHINE

	SIM2x00	●		●	●	●	●			●	●	●	●
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1) In combination with a Sensor Integration Machine unit (e.g. SIM2500 Nova)
2) Requires integration-software
3) CX-hardware versions using SOPAS ET software
4) AP-hardware versions
5) Depends on the infrared remission properties of the target object
6) With internal illumination, can be extended to longer distances when using external illumination
7) Currently not supported by SICK Nova

TECHNOLOGY		COMMUNICATION INTERFACE				SENSOR RESOLUTION	WORKING DISTANCE	PAGE
Snapshot (2D/3D)	3D laser triangulation	Serial	Ethernet TCP/IP	Fieldbuses	GigE Vision			

●		● 7)	●	●		1.2 Mpixel	50 mm ... 300 mm ^{d)}	→ 10
●		● 7)	●	●		1.3 Mpixel	70 mm ... 1,500 mm	→ 10
●		● 7)	●	●		1.3 Mpixel ... 1.9 Mpixel	Depending on optics	→ 11
●		● 7)	●	●		1.3 Mpixel ... 5.1 Mpixel	Depending on optics	→ 11
●		● 7)	●	●		5 Mpixel ... 12 Mpixel	Depending on optics	→ 11
●			●		●	1.6 Mpixel ... 5.0 Mpixel	Depending on optics	→ 12
●			●		●	1.6 Mpixel ... 12.3 Mpixel	Depending on optics	→ 12

	●	●	●			1536 datapoint/profile	141 mm ... 1091 mm	→ 14
	●		●		●	3200 datapoint/profile	Depending on optics	→ 14
	●		●		●	3200 datapoint/profile	46,7 mm ... 1445 mm	→ 13
●			●			640 px x 512 px	0.5 m ... 6.5 m 0.5 m ... 65 m	→ 15
●			●			512 px x 424 px	≤ 16 m ⁵⁾	→ 15
●			●		●	1024 x 576 px	0.3 m ... 37m	→ 15

		● 7)	●	●	●			→ 13
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InspectorP61x

Powerful 2D vision sensor for tight spaces



InspectorP62x

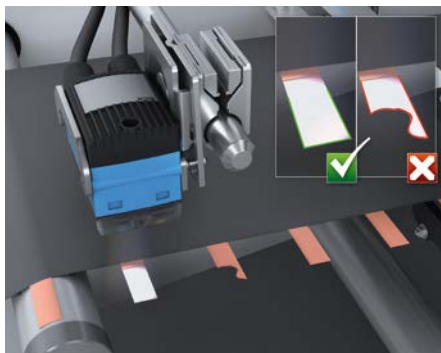
All-in-one vision sensor with electric focus and rugged housing

POSSIBLE FIELDS OF APPLICATION

- Inline quality control for electronics, packaging, and automotive industries
 - Manufacturing and assembly verification
 - Part localization and measurement
 - Detail inspection, counting, and measurement
 - Perfect for robot end-of-arm mounting
 - Print and label inspection with code reading and OCV
 - Track and trace by code reading and OCR
- Inline quality control in the production of parts, assemblies or products
 - Inspection and measurement of details
 - Print and label inspection with code reading and OCV
 - Track and trace by code reading and OCR
 - Other SensorApps, systems or custom development for additional tasks, e.g.:
 - Fine positioning of stacker cranes

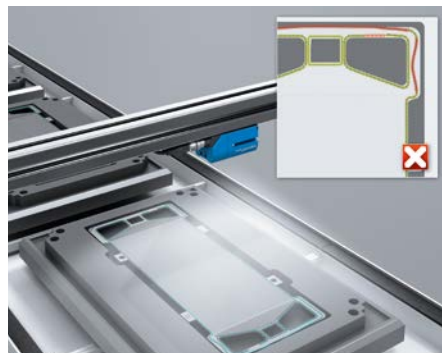
EXAMPLE APPLICATION

BATTERY



Quality control of electrode tabs

ELECTRONICS



Verifying the seal

DETAILED INFORMATION

Learn more
sick.com/InspectorP61x



Learn more
sick.com/InspectorP62x





InspectorP63x

All-in-one vision sensor with flexible optics and rugged housing

- Robot picking
- Precise part positioning
- Quality inspection of shapes and dimensions
- Defect and part presence detection
- Calibrated, high-accuracy measurements
- Track and trace by code reading and OCR
- Print and label inspection with code reading and OCV
- Object sorting

Inspector83x

AI-powered quality control made stress-free

- Complex and unpredictable defect detection
- Sorting of parts, products or packages
- Inline, high-speed quality control in the production of parts, assembly, or finalized products
- Manufacturing and assembly verification
- Track and trace by OCR and code reading
- OCR/OCV and quality inspection in packaging lines

Inspector85x

Focusing on the big picture

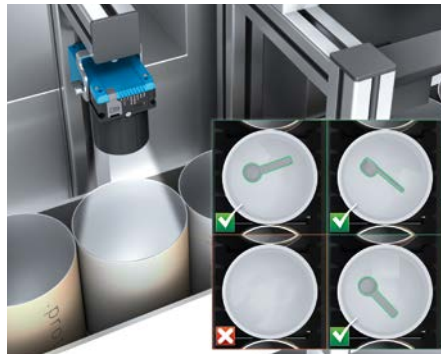
- In-line quality inspection in automotive and white goods production lines that require large field of view
- OCR/OCV and quality inspection in packaging lines for consumer goods
- Solder inspection of electronics and automotive components
- Complex and unpredictable defect detection

RUBBER AND PLASTICS



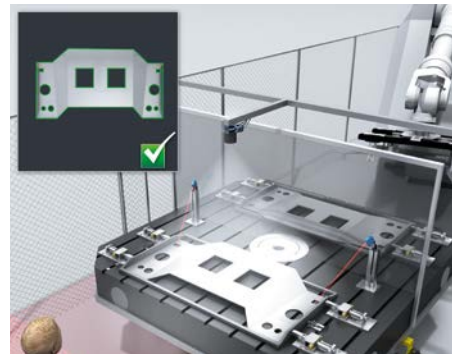
Monitoring of the injection mold

CONSUMER GOODS



Completeness verification

AUTOMOTIVE



Type check for bodyparts

Learn more
sick.com/InspectorP63x



Learn more
sick.com/Inspector83x



Learn more
sick.com/Inspector85x





picoCam2

Ultra compact industrial streaming cameras in accordance with GigE vision standard



midiCam2

Rugged and high-resolution streaming camera in accordance with GigE vision standard

POSSIBLE FIELDS OF APPLICATION

- Fast in-line quality control for consumer goods, automotive and electronics industries
- Manufacturing and assembly verification
- Part localization and measurement
- Detail inspection, counting and measurement
- Color verification and sorting
- Challenging classification and anomaly applications

- Fast in-line quality control for consumer goods, automotive and electronics industries
- Manufacturing and assembly verification
- Part localization and measurement
- Detail inspection, counting and measurement
- Color verification and sorting
- Challenging classification and anomaly applications

EXAMPLE APPLICATION

AUTOMOTIVE



Inline quality control

CONSUMER GOODS



Inline quality control

DETAILED INFORMATION

Learn more
sick.com/picoCam2



Learn more
sick.com/midiCam2





SIM2x00 Nova

Combining SIM2x00 with picoCam2, midiCam2, Ruler3000, Ranger3 or LMS4000 enables high-performance, customizable 2D and 3D machine vision solutions easily solved with SICK Nova

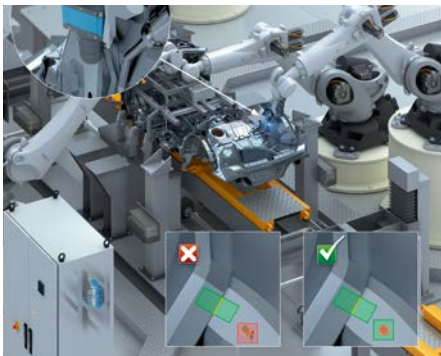
- Fast in-line quality control for consumer goods, automotive and electronics industries
- Manufacturing and assembly verification
- Part localization and measurement
- Detail inspection, counting and measurement
- Color verification and sorting
- Challenging classification and anomaly applications

Ruler3000

The fast way to high-performance 3D

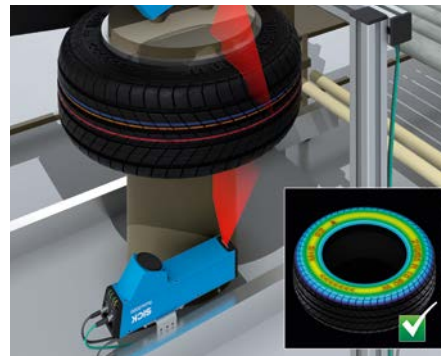
- Assembly inspection of tablets and cell phones in the electronics industry
- Quality control of tires
- Weld seam inspection of batteries
- Packaging and in-line food quality control
- Inspection of boards and 3D log measurement in the timber industry
- Article identification in logistics
- Robot guided inspection

AUTOMOTIVE



Measuring the welding edge and spot welding inspection

AUTOMOTIVE



Lettering inspection/side walls of tires

Learn more

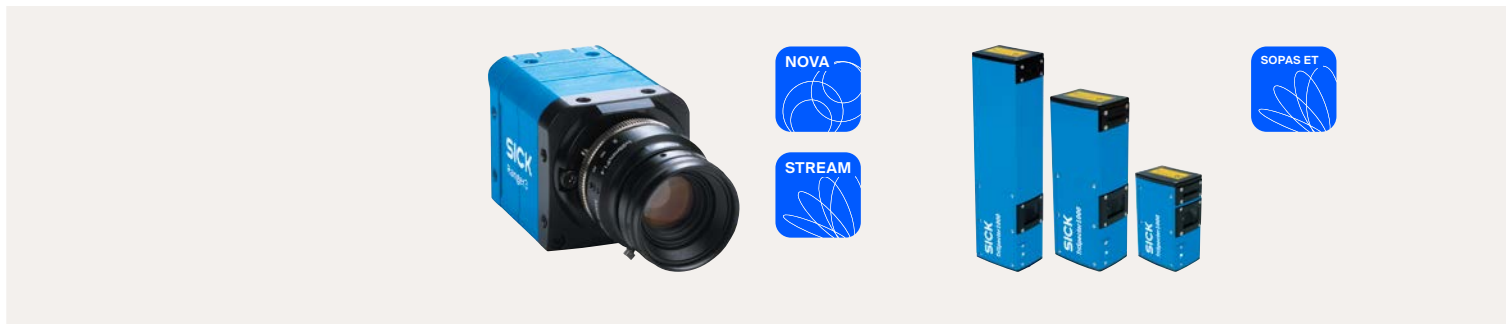
sick.com/SIM2x00



Learn more

sick.com/Ruler3000





Ranger3

Superior 3D performance in a small package powered by ROCC technology

TriSpector1000

Configurable, intuitive, and reliable 3D inspection of moving parts

POSSIBLE FIELDS OF APPLICATION

- Electronic component and PCB inspection
 - Quality control of tires
 - High-speed railway and road surface inspection
 - Dimensioning of tubes in the metal industry
 - Packaging and in-line food quality control
 - Quality inspection and classification in the wood and furniture industry
- Quality control in consumer goods; volume and thickness measuring, counting and positioning of objects
 - Box integrity including content verification, completeness and emptiness check
 - Product dimensioning in food processing

EXAMPLE APPLICATION

RAIL



Inspection of train components

FOOD AND BEVERAGE



Box content verification

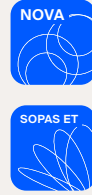
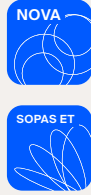
DETAILED INFORMATION

Learn more
sick.com/Ranger3



Learn more
sick.com/Trispector1000





Visionary-S

3D snapshot – colorful perspectives for precise factory automation

- Bin picking
- Navigation and positioning in robotics
- Quality control, e.g. of secondary packaging
- Palletizing and depalletizing
- Dimensioning of goods
- Verification, e.g. empty container detection at airports
- Checking the packaging content
- Checking the manufacturing and assembly
- Completeness or empty checking

Visionary-T Mini

3D machine vision has never been easier

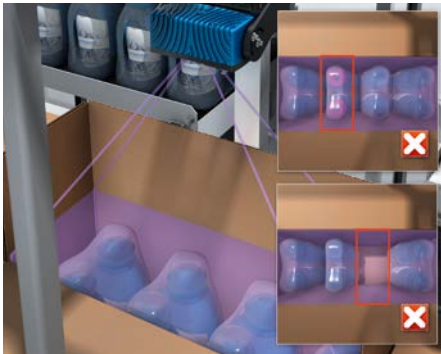
- Object detection
- Navigation
- Palletizing and depalletizing
- Measurement and volume detection
- Positioning
- Gesture control
- Area monitoring
- Checking the packaging content
- Checking the manufacturing and assembly
- Completeness or empty checking
- Fill level monitoring

Visionary-B Two

Rugged and smart - 3D snapshot solution for harsh outdoor environment

- Object detection
- Navigation
- Positioning
- Area surveillance
- Fill level monitoring

CONSUMER GOODS



Presence detection

LOGISTICS



Fill level monitoring

MINING



People Detection

Learn more
sick.com/Visionary-S



Learn more
sick.com/Visionary-T_Mini



Learn more
sick.com/Visionary-B_Two





APPLICATION SPECIFIC SOFTWARE

Learn more

→ sick.com/Application_Software

Application specific software from SICK extends the functionality of sensors and systems for complex automation tasks. It either solves applications out-of-the-box or can easily be configured to individual solutions even powered by AI. This allows customers to implement their applications efficiently - without any development effort on their part.

3DBP

3D BELT PICK

CONVEYOR BELT PICKING MADE SIMPLE IN ALL DIMENSIONS

- Provides 3D coordinates and measurements of products on belt
- Easy integration with several robot brands
- Running stand-alone on TrispectorP1000 and on Ruler3000 bundle

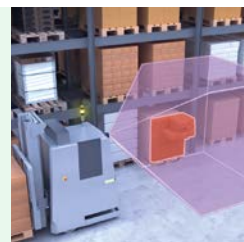


3DOD

3D OBJECT DETECTION

FOR COLLISION AVOIDANCE APPLICATIONS

- Flexible collision avoidance with up to 16 switchable detection fields and freely configurable object sizes
- Prime Version with ground filter for e.g. forklift fork detection when left on the ground
- Easy to integrate via I/O interface or via Ethernet with REST API

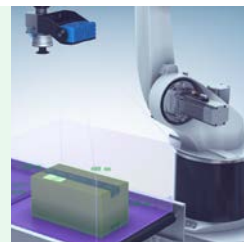


STADIM

STATIC PACKAGE DIMENSIONING

SENSORAPP TO ESTIMATE DIMENSION (LENGTH, WIDTH, HEIGHT) OF THE SMALLEST ENCLOSING CUBE AROUND ONE OR A GROUP OF OBJECTS

- Stand-alone solution - no additional computer needed
- Access to dimensioning result (L,W,H) via REST API
- SensorApp can be loaded to Visionary-S AP devices

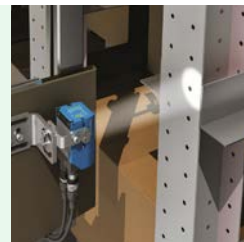


RFP

INSPECTORP RACK FINE POSITIONING

FINE POSITIONING SOLUTION FOR SINGLE- AND DOUBLE-DEEP RACKING

- Single- and double-depth rack storage using one device
- User-friendly web interface
- Pre-assembled, pre-focused and pre-configured



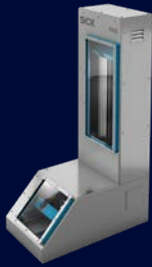
TTVS

TRACK AND TRACE VISION SOFTWARE

EFFICIENCY IN SORTING PROCESSES THANKS TO SMART, IMAGE-BASED OBJECT CHARACTERIZATION

- Optimizes the sorting process of various objects and helps with correct handling
- Real-time results without additional hardware - runs directly on the camera
- Plug-and-play software - no previous knowledge needed





SYSTEM SOLUTIONS

Learn more and find all available system solutions

→ sick.com/Systems

System solutions are the right choice when it comes to automating and optimizing large-scale industrial applications with stringent requirements on efficiency and resilience. With a broad range of intelligent sensors, software, digital services, and service offerings available, it is possible to find a suitable solution for every requirement.

PLB

FLEXIBLE PART LOCALIZATION FOR BIN PICKING AND DEPALLETIZING

- Several localization algorithms based on CAD, geometric part features, and AI
- Selection of various pre-calibrated 3D cameras for immediate use
- Easy integration with any robot model



PALLOC

EASY DE-PALLETIZING OF SACKS AND BAGS WITH 3D COLOR VISION BASED ON DEEP LEARNING

- Localization function based on deep learning ensures reliable depalletizing
- Robot software interface packages for many brands
- Powerful embedded processing ensures high picking speed



PLOC2D

EASY TO USE AND FLEXIBLE PART LOCALIZATION SENSOR SYSTEM

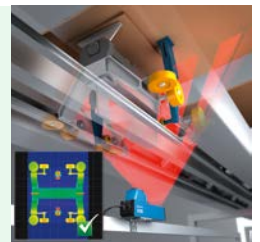
- 2D part localization
- Stand-alone sensor with web-based user interface
- Easy integration with robots



SORTER INSPECTION SYSTEM

CONTINUOUS MONITORING FOR OPTIMIZED MAINTENANCE AND EARLY DEFECT DETECTION

- Fewer planned and unplanned downtimes
- Avoidance of consequential damage due to overlooked defects
- High system availability and lower maintenance effort and costs



PALLET INTEGRITY INSPECTION SYSTEM

CONTACTLESS PALLET CONTROL FOR FAST AND RELIABLE DETECTION OF VARIOUS DEFECTS

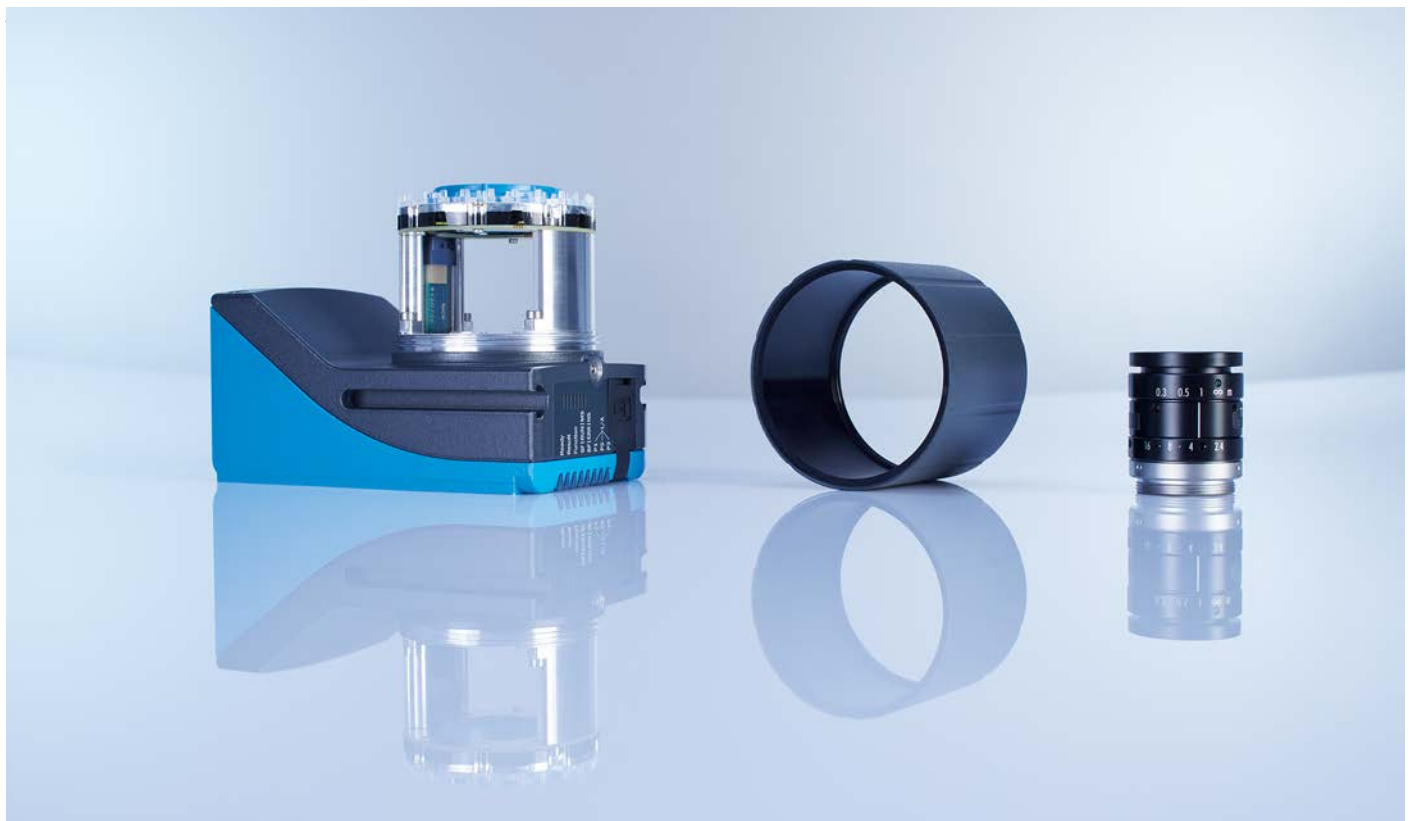
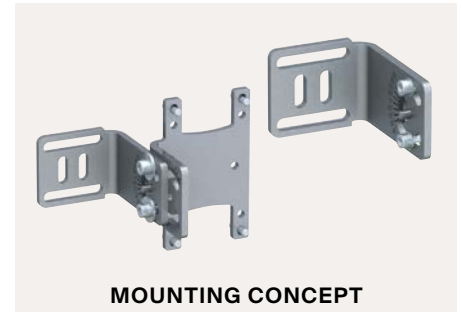
- High-precision detection of missing parts and damage on pallets
- Instant knowledge of the status of each pallet in real time
- Seamless integration into existing plants



ACCESSORIES FOR THE MACHINE VISION PORTFOLIO

Learn more
→ sick.com/Accessories

Different needs require different solutions. Fitting the respective customer requirements on site and exactly matched to the products, these accessories enable quick and easy sensor integration into existing systems. This makes applications efficient and reliable.



WORKING WITH SICK IN A DIGITAL WORLD

MAKING YOUR DIGITAL BUSINESS ENVIRONMENT COMFORTABLE

FIND A SUITABLE SOLUTION IN NEXT TO NO TIME

- Online product catalog
- Application Solver
- Online configurators and selectors

MY SICK IS YOUR PERSONAL SELF-SERVICE PORTAL

- Open around the clock
- Clear product information
- Company-specific price conditions
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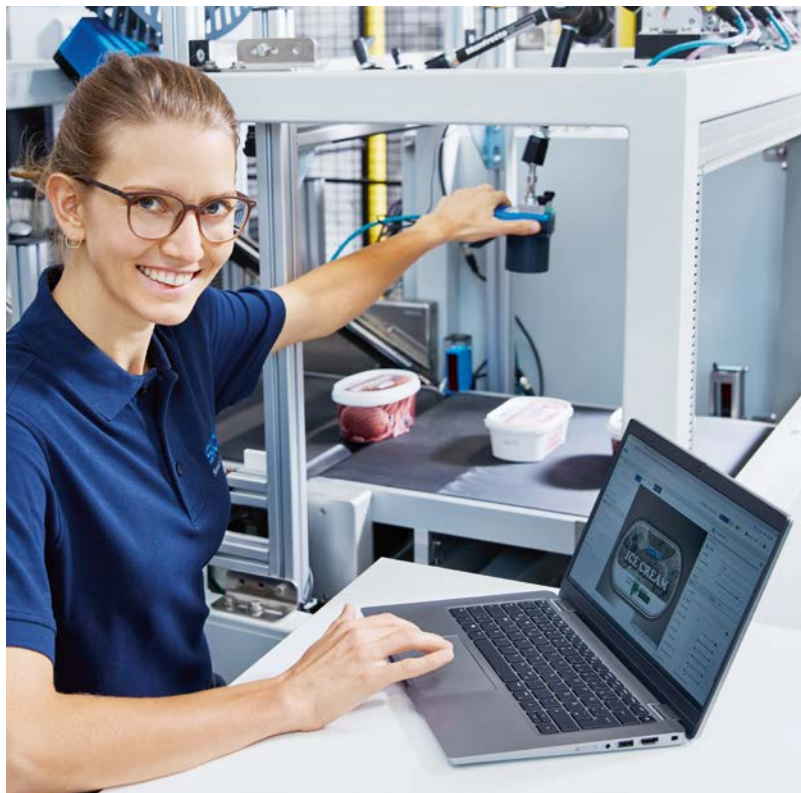
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